Inverse Heat Modelling and Scientific Diagnostics

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1 Formulation

Model. $u_t = \partial_x[a(x)u_x]$, $a = a_0 e^{\beta(2x-1)}$, $u(0,t) = g(t)$, $-a(1)u_x(1,t) = h[u(1,t) - T_a(t)]$; readings $y = u(x_s,t) + b_s + \varepsilon$, $b_{\text{ref}} = 0$. a_0 is the mid-rod conductivity, $e^{2\beta}$ the right/left ratio; a heater clamps x=0 and Newton cooling acts at x=1. Heat spreads only $\sqrt{at} \approx 0.3$–0.4 in an experiment, so only the right-most sensors see the Robin end: h is expected to be the hardest parameter, and it trades off with conduction and offsets.

Inverse problem. $\theta = (a_0, \beta, h, b_s)$ within the brief's bounds; minimise $\frac{1}{2}\sum_k[(u_k(\theta) + b_{s(k)} - y_k)/\sigma_k]^2$, the Gaussian likelihood with the supplied noise sd. Offsets are persistent per sensor and enter linearly; the reference sensor fixes the level. Uncertainty: Laplace covariance $(J^\top J)^{-1}$, 95% Wald intervals (local, conditional on the model). Assumptions — independent Gaussian noise with known sd, correct model form, piecewise-linear inputs — are each tested by a guard (§5).

Data. All data are synthetic, generated to the brief's schema with truth kept in a separate file read only for scoring; results on them are marked[SYN]. Cleaning is explicit and logged per row: 10 `bad`, 8 conflicting repeated-ID (all copies), 6 duplicate, 5 missing and 2 out-of-domain rows dropped; 6 `suspect` kept. Fit E1–E3, validate E4 (labels never reach fitting), predict E5 from its inputs only. Methods were frozen at `9abde6c`; estimates are bit-identical across the three evaluation runs.

2 Stage 1 — reference solver and inference

Cell-centred finite volumes (conservative), Robin end as half-cell conduction in series with h, Crank–Nicolson with Rannacher start-up. **Verification**, independent of the inverse code: manufactured solution, order 1.993–1.999 (Fig. 2a), RMSE 3.3e-05 at the production grid (100×400) vs noise $\approx 10^{-2}$; closed-form steady state 2.9e-06; energy balance 2e-16; maximum principle. Data are generated on a 4× finer grid (no inverse crime). **Fitting:** bounded trust-region least squares, 4 Latin-hypercube starts. Known-parameter recovery is tested on separate synthetic data; coverage over 20 datasets is in §4.

Table 1: Public bundle[SYN]. Truth a_0=0.104, β= − 0.361, h=0.322. Intervals 95% Laplace. "noisy" = vs readings, "latent" = vs noise-free u at E5. PINN: mean of 3 seeds, [min,max] (a spread, not an interval).

	FV, variable $a(x)$ (primary)	FV, constant a	inverse PINN
a_0	0.107 [0.100, 0.113]	0.120 [0.109, 0.131]	0.094 [0.091, 0.096]
β	-0.405 [-0.471, -0.339]	0 (fixed)	-0.426 [-0.444, -0.413]
h	0.330 [0.272, 0.388]	0.278 [0.215, 0.340]	0.174 [0.158, 0.185]
χ^2_ν (train)	1.12	1.48	—
train RMSE (noisy)	0.0135	0.0158	0.0137
val E4 RMSE / 95% coverage	0.0124 / 0.96	0.0142 / 0.92	0.0164 / —
test E5 RMSE noisy / latent	0.0131 / 0.0012	0.0139 / 0.0084	0.0169 / 0.0144
test latent-interval coverage	0.99	0.33	—
fit time (laptop CPU)	6.1 s	5.6 s	243 s/seed + 59 s adapt.
energy-balance defect	5e-16	—	0.022 (relative)
status (automatic)	ok	weak_evidence	3/3 weak_evidence

Result. The variable model recovers all three parameters inside their intervals (χ^2_ν=1.12). The constant baseline is the cautionary case: almost the same error against noisy readings (0.0139 vs 0.0131) but 7× the latent error (0.0084) and latent intervals covering the truth only 0.33 of the time — offsets absorb the missing gradient. The nested test rejects it ($\Delta\chi^2$=150; Fig. 1).

3 Stage 2 — inverse PINN

Design. One 4×48 tanh MLP per training experiment (smooth second derivatives; each experiment has its own forcing, the physics is shared); a_0, β, h bounded by sigmoids, offsets by 0.03 tanh. Losses $L_{\text{obs}}, L_{\text{pde}}, L_{\text{ic}}, L_{\text{bc}}$ are in noise units (divided by the median sensor variance, weights 1), so violating physics by one noise level costs about one misfitted reading. L_{pde} uses the conservative operator by autograd: flux = $a(x)u_x$, then ∂_xflux, so $a'(x)u_x$ appears automatically (tested against the hand-derived source to 10^{-10}). 2000 uniform collocation points

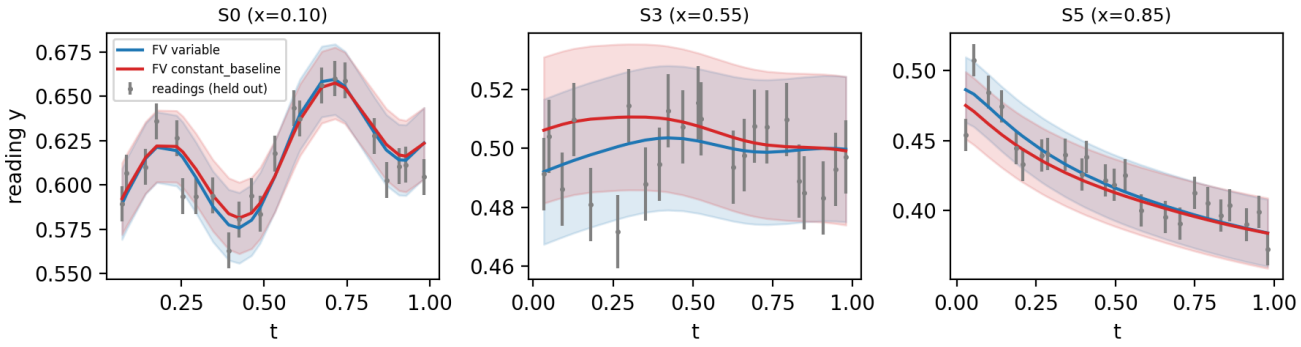


Figure 1: Held-out E4[SYN]: readings vs FV variable (blue) and constant (red) predictions with 95% reading intervals. Both look plausible on readings.

per experiment, resampled every 250 Adam steps; then L-BFGS on a fixed set. Physics is frozen for 1500 warm-up steps. Unseen experiments: physics-only adaptation, 59 s/seed.

Development (separate dev bundle, one change per trial, `docs/evidence`): no warm-up → h collapsed onto its bound; warm-up → a_0, β right, h 3× low; w_{bc}=10 → "BC weight pins h" refuted; more steps marginal; w_{pde}=10 worse. Mechanism: soft constraints leave about one noise level of slack between the last sensor (x=0.85) and the Robin end, enough for the field to bend so a wrong h satisfies the boundary condition. Tuning stopped there.

Result. The PINN fits the readings (train RMSE 0.0137) but its parameters are off: h by 46%, a_0 by 10%; only β agrees with FV. Its seed spread is not uncertainty, and agreement with an FV field at its own parameters (RMSE 0.0029) is consistency, not accuracy. An exact-physics likelihood-ratio check places every seed outside the FV fit's 99.9% region ($\Delta\chi^2$ 37–58 vs 26.1), so all are reported `weak_evidence`. The PINN does not beat the numerical baseline here; FV is primary.

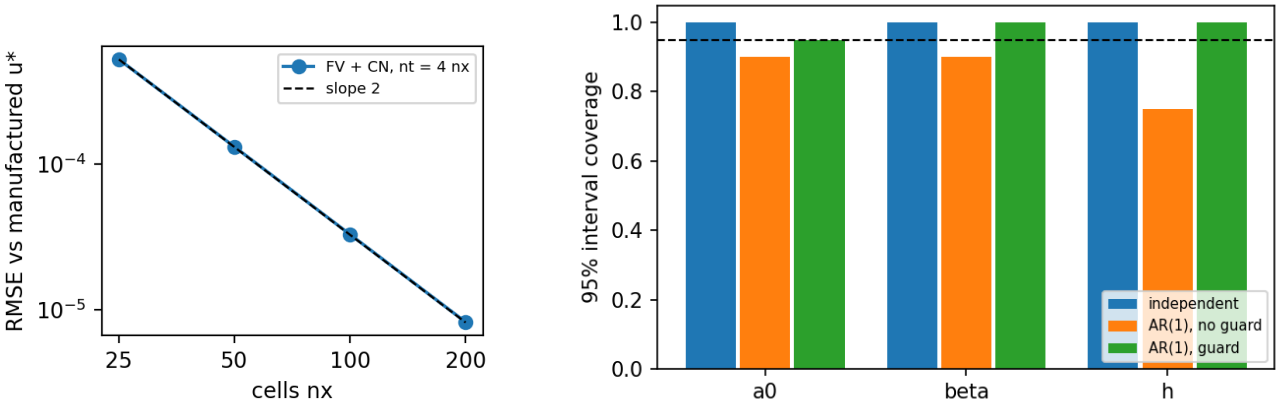


Figure 2: (a) Manufactured-solution refinement: slope 2. (b) Interval coverage over 20 datasets[SYN]: independent noise; AR(1) noise without and with the autocorrelation guard.

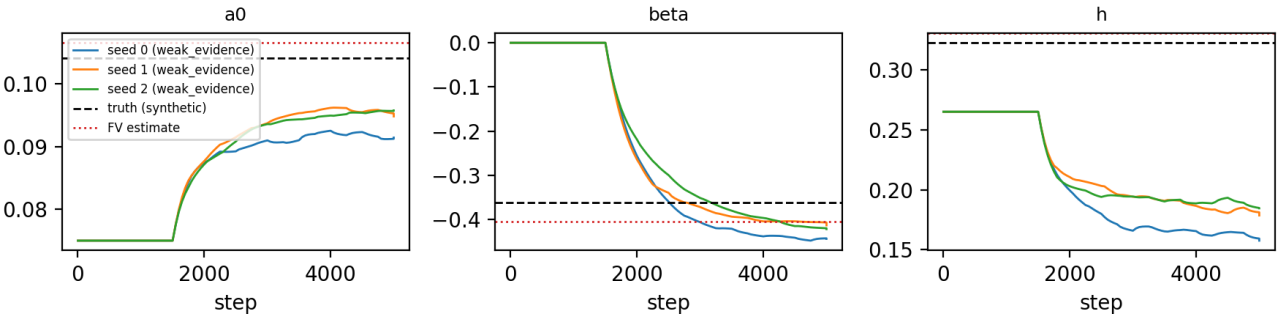


Figure 3: PINN parameters during training, public bundle[SYN], 3 seeds (physics frozen for the first 1500 steps). β settles near FV (dotted); a_0 ends 10% low and h far below truth (dashed).

4 Stage 3 — investigation

Uncertainty (Fig. 2b). Coverage of 95% intervals on 20 independent synthetic datasets: 20, 20, 20 of 20 for a_0, β, h (Wilson lower bound 0.84; small n, a sanity check). Data uncertainty alone (20 noise redraws): empirical/Laplace sd 1.25, 1.24, 1.29 — mildly optimistic, not conclusive at n=20. Seed variability is separate: zero for FV (multistarts agree), the Table 1 spread for the PINN. With AR(1) noise (ρ=0.6) plain intervals cover 18, 18, 15 of 20; the guard flags all 20 and widens by $(1+\hat{\rho})/(1-\hat{\rho})$, restoring 19, 20, 20; it flags 0 independent-noise datasets.

Controlled experiments (Fig. 4). One change each, hypotheses stated in code beforehand, fixed evaluation (held-out latent RMSE). Removing sensors at $x \geq 0.7$ or keeping only $t \leq 0.3$ widens the h interval 5.3×; early times also widen β; dropping left sensors mainly widens β. Assuming constant a gives 9× the held-out latent error; forcing offsets to zero pushes β to its bound (0.700).

Experimental design. A new sensor at x=0.95 vs the same 24 readings per experiment on existing sensors. Rule, fixed before truth: pick the design with the smaller predicted sd of h at a pilot fit. It chose the new sensor in 10/10 datasets and narrowed the h interval in 10/10 (width 0.0376 vs 0.0487), but $|h|$ error did not measurably differ (0.0078 vs 0.0072): a precision gain, not a demonstrated accuracy gain.

The three questions. *Can bias mask wrong physics?* Yes: with β=0.45 the constant-a model fits at χ^2_ν=1.17 (limit 1.20, accepted) with 9× the latent error; only model comparison or held-out prediction exposes it. *Which parameters become weak?* h first (right-end or late data removed), then β (early times); a_0 stays determined. *Which measurement helps?* A sensor near the Robin end: it sees the boundary gradient directly and breaks the h/conduction/offset trade-off.

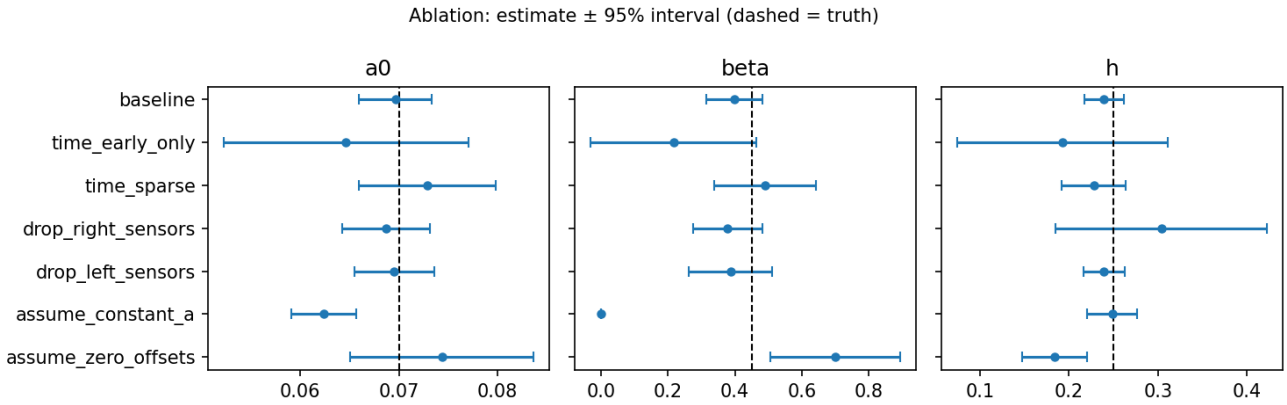


Figure 4: Controlled experiments^{SYN}: estimate ±95% interval under each change (dashed: truth).

5 Stage 4 — reliability, diagnosis, recovery

Automatic status. Every result is `ok`, `weak_evidence` or `failed` from rules in code that never use truth: $\chi^2_\nu > 1+3\sqrt{2/\nu}$; positive residual autocorrelation (z>3); held-out predictive χ^2; nested test constant vs variable a (p=10^{-3}); interval wider than 50% of the range ⇒ estimate withheld; estimate on a bound; multistart disagreement; PINN exact-physics check; fault scan. False-alarm rates were measured on correct models.

Diagnosis. Rules frozen in `configs/diagnosis_rules.yaml`. Observations (residual trends, χ^2, early/late parameter shift) are reported separately from causal hypotheses, each a one-parameter extension: drift of each sensor, a Gaussian heat source (centre scanned), a bias in the recorded g. Evidence is the χ^2 drop; the extra parameter ± sd is the sensitivity. Threshold 15.1 (Bonferroni, 10 hypotheses, family p=10^{-3}); on 13 fault-free bundles the largest drop is 6.8 (0 false alarms); 12/12 known faults correct; fault-size sweep 13/22 detected, 0 named wrongly. Undetected small faults still bias β by up to 0.09. **Blind test:** rules committed, then a sealed bundle with a randomly drawn fault and its truth (hash in the manifest), then the verdict, then unsealing; re-running with truth reads blocked gives the same verdict. Verdict *suspected:heat source* ($\Delta\chi^2$=38, Q=0.051 ± 0.009); truth: heat source, Q=0.082 at x=0.78 (integrated heat within 6%). The base fit there said `ok` with h=0.178 (true 0.288), so the fault scan now runs inside `evaluate` (public bundle: no fault detected).

Recovery and visibility. Checkpoints carry a fingerprint (config + data hash). A run killed mid-training and resumed matches the uninterrupted run exactly (difference 0); a checkpoint from another config is refused; a truncated file is skipped; an injected non-finite loss is logged, rolled back and the learning rate halved. Every result logs data hash, config, seed, git commit, hardware, versions and wall time.

A misleading result. In the first evaluation a PINN seed passed the old check with h=0.185 (true 0.322):

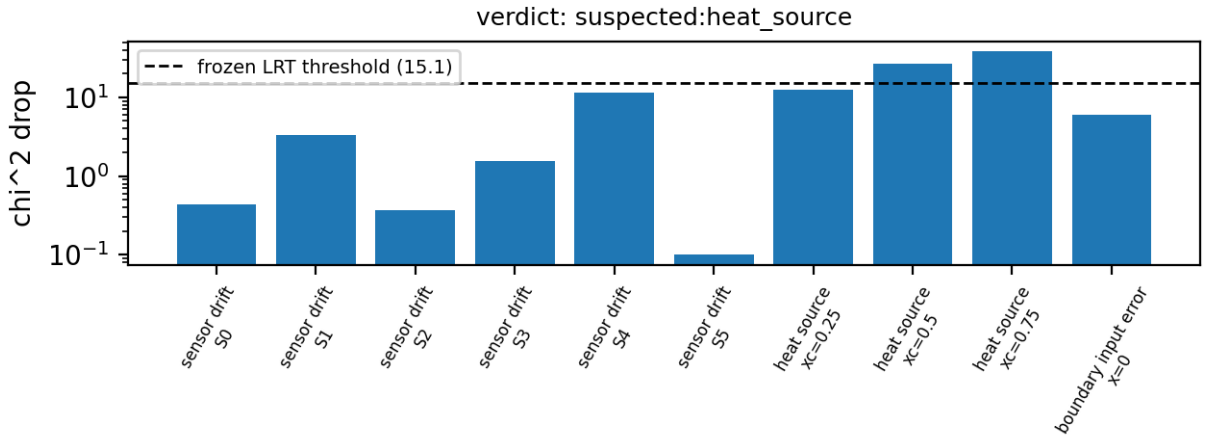


Figure 5: Sealed blind test: χ^2 drop per fault hypothesis vs the frozen threshold 15.1; only the heat source clears it with margin.

χ^2_ν=1.2077 vs limit 1.2078. A wrong h moves χ^2_ν by only 0.09, so an absolute test has little power; the replacement compares against the FV optimum on the same data and rejects it clearly. Testing also found six input-handling bugs, all fixed.

Compute. Laptop i7-1165G7, 16 GB, no GPU: FV fit 6.1 s; PINN 243 s/seed + 59 s adaptation; public evaluation 16 min; CPU smoke run 169 s (PASS, budget 10 min) on a 2-vCPU cloud runner; 58 tests pass.

6 Claims, limitations, time

Supported^{syn}: FV recovery with calibrated intervals; h is the weakly identified parameter and a Robin-end sensor improves its precision; guards detect dependent noise, wrong diffusivity form, PINN parameter inconsistency and three fault types above measured sizes. **Not supported:** anything on real data; faults below detection bias β; Laplace intervals are local; the PINN's h; design accuracy gain; coverage from only 20 datasets. **Time spent:** ≈7 h build, experiments and report; cloud smoke run, fact-check of every number and fixes; author review and walkthrough preparation.

Appendix — AI assistance and verification

How AI was used. Claude was used as a pair-programmer throughout: drafting code, running experiments and drafting this report. I directed the plan, decisions and trade-offs; decisions such as stopping PINN tuning, adding guards and tightening thresholds were discussed and approved explicitly.

How the output was verified — by evidence, not trust.

- *Independent references:* the solver against a manufactured solution, a closed-form steady state, energy conservation and the maximum principle; the PINN operator against the same hand-derived source.
- *Known truth:* all studies use synthetic data with truth stored separately and read only for scoring; a file guard proves fit/predict never open it; corrupting validation labels leaves training estimates bit-identical.
- *Pre-registration:* diagnosis rules committed before the sealed blind test, methods before the final evaluation; rule changes (one-sided autocorrelation test, Bonferroni threshold, stronger PINN check after the first evaluation run) are declared with their evidence.
- *Adversarial testing:* 58 tests (metamorphic shift/scale invariance, renamed IDs and layouts, row order, edge-of-box truths, malformed inputs, CLI; autograd operator, Jacobian, decision branches). They found six real bugs, one wrong test premise and a dependency-pin gap on a fresh clone.
- *AI errors caught:* an ad-hoc threshold, a guard with too little power, a premature claim that a status rule was protective, a mis-stated clock time, and a figure drawing the superseded diagnosis threshold. Each was challenged, re-measured and corrected.
- *No hand-typed results:* every number here is generated by `report/make_numbers.py` from the result files.